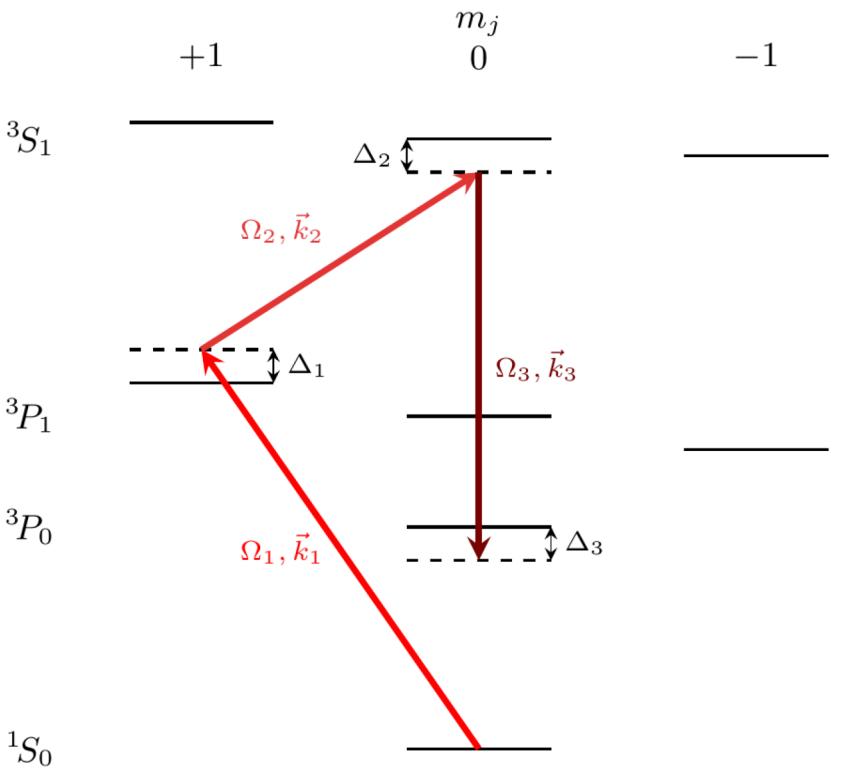
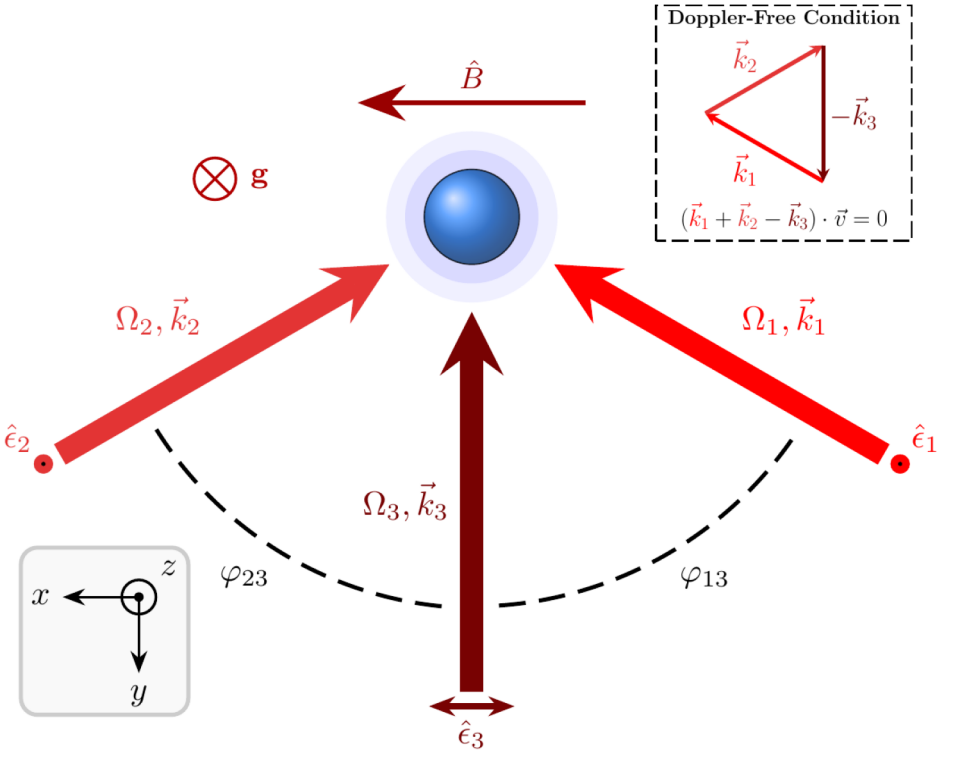
a)


m_j
$+1$
0
-1
$^3S_1$
$^3P_1$
$^3P_0$
$^1S_0$
Δ_2
Δ_1
Δ_3
$\Omega_2, \vec{k}_2$
$\Omega_3, \vec{k}_3$
$\Omega_1, \vec{k}_1$

b)


Doppler-Free Condition
$\vec{k}_2$
$-\vec{k}_3$
$\vec{k}_1$
$(\vec{k}_1 + \vec{k}_2 - \vec{k}_3) \cdot \vec{v} = 0$
$\hat{B}$
$\mathbf{g}$
$\Omega_2, \vec{k}_2$
$\Omega_1, \vec{k}_1$
$\Omega_3, \vec{k}_3$
$\hat{\epsilon}_2$
$\hat{\epsilon}_1$
$\hat{\epsilon}_3$
φ_{23}
φ_{13}
x
y
z